AAN 500 Final Assessment

FlavorWave QSR Digital Transformation

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Background & Current Situation

FlavorWave is a fast-casual restaurant chain that invested in AI and analytics to improve inventory management and customer engagement.

50
Restaurant locations

$10M
Invested in AI and analytics

18 months
Into the two-year initiative

So far neither initiative has achieved the expected results. With six months remaining, FlavorWave needs to identify the causes of these problems and determine how to improve its AI initiatives before making further investments.

Data Quality & System Integration

FlavorWave has several data quality issues that could affect its AI predictions, including:

- 2 of 20 inventory records had negative stock quantities, indicating possible data errors.
- There are potential duplicate sales records.
- Some customer information is missing, and employee performance scores fall outside the expected range.

Additionally, older systems may not share data properly, making inventory predictions less accurate.

Item ID	Item Name	Category	Current Stock
I001	Beef Patty	Meat	500
I002	Lettuce	Produce	-50
I003	Buns	Bakery	1000
I004	Tomatoes	Produce	300
I005	Cheese	Dairy	200
I006	Chicken Breast	Meat	400
I007	Whole Wheat Wrap	Bakery	600
I008	Avocado	Produce	150
I009	Vegan Patty	Meat Substitute	200
I010	Spinach	Produce	180
I011	Gluten-Free Bun	Bakery	250
I012	Beef Patty	Meat	500
I013	Fries	Frozen	400
I014	Mixed Greens	Produce	220
I015	Turkey Slice	Meat	300
I016	Quinoa	Grains	150
I017	Egg	Protein	500
I018	Almond Milk	Dairy Alternative	-180
I019	Sweet Potato	Produce	250

Image of negative stock quantities

Root Causes of FlavorWave's Challenges

Lack of clear leadership

The previous AI leader left without a clear implementation plan, which can leave employees unsure of their responsibilities.

Employee resistance

Employees may be hesitant to use AI because of concerns about job security and increased workloads.

Limited training & technical skills

Limited training and technical expertise may make it difficult to use and maintain the AI systems.

Proposed Analytical Model

FlavorWave could use a demand forecasting model to predict how much of each ingredient its restaurants will need.

The model would use past sales and recipe information to estimate demand. It would also account for current inventory and upcoming deliveries to help managers decide how much to order.

To test the model, FlavorWave can compare its predictions with actual ingredient usage.







Model development would follow the CRISP-DM framework (Vardhan, 2023).

Ethical Considerations & Industry Best Practices

Ethical Considerations

- Employees may worry about job security and how AI will affect their responsibilities.
- Customer information must be protected and handled in accordance with privacy laws.
- AI generated discounts and promotions could unintentionally favor certain customer group.

Recommended Practices

To address these concerns and follow industry best practices, FlavorWave should:

- Involve employees in testing and provide training before introducing new tools.
- Limit access to customer information and explain how it is used.
- Regularly review AI recommendations and assign responsibility for addressing problems (Papagiannidis et al., 2025).

Implementation Plan & Business Value

FlavorWave should use the remaining six months to address its current challenges and test the inventory forecasting model before expanding it.

First 30 days:

Review data quality, existing systems, & employee feedback

→

Days 31-60

Correct data issues, prepare the model, & train employees

→

Days 61-90

Test the model at a few restaurants & review its predictions

→

Months 4-6

Evaluate results & decide whether to expand

To measure success, FlavorWave can compare food waste, ingredient shortages, and inventory costs between restaurants using the model and those that aren't. Reducing waste and avoiding shortages could lower costs and prevent lost sales, helping the company see whether its AI investment is providing real business value.

References

FlavorWave QSR dataset [Data set]. (n.d.). Google Sheets. https://docs.google.com/spreadsheets/d/1ejoxRJ5q-L01o4dSVlYAO1c00x1OJ21DKI7xUW7uvMI/edit?gid=2079607299

Papagiannidis, E., Mikalef, P., & Conboy, K. (2025). *Responsible artificial intelligence governance: A review and research framework*. The Journal of Strategic Information Systems, 34(2), Article 101885. https://doi.org/10.1016/j.jsis.2024.101885

Vardhan, H. (2023, October 29). Data science process frameworks. https://harshvardhan.blog/data-science-process-frameworks